MTH240

Examination - July Semester 2025

Engineering Mathematics I

Friday, 14 November 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Consider the following system of linear equations:

$$\begin{aligned} x + 2y - 4z + 3r - s &= 0 \\ x + 2y - 2z + 2r + s &= 0 \\ 2x + 4y - 2z + 3r + 4s &= 0\,. \end{aligned}$$

Question 1a

Expressing the system of linear equations above as $AX = 0$ where

$$X = \begin{bmatrix} x \\ y \\ z \\ r \\ s \end{bmatrix}$$

and 0 denotes the 3×1 zero matrix, write down the matrix A .

(5 marks)

Question 1b

Solve the above system linear equations.

(7 marks)

Question 1c

Compute bases for the row space and column space of the matrix A in Question 1a.

(8 marks)

Question 2

Let

$$M = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Question 2a

Evaluate the eigenvalues and eigenvectors of M.

(15 marks)

Question 2b

Is the matrix diagonalizable? Justify your answer.

(5 marks)

Question 2c

Consider the subspace U of $\mathbb{R}^4$ spanned by the vectors:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \; v_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}, \; v_3 = \begin{bmatrix} 1 \\ 2 \\ -4 \\ -3 \end{bmatrix}.$$

Use the Gram-Schmidt process to find an orthonormal basis of U.

(10 marks)

Question 3

A hospital receives emergency calls, which follow a Poisson distribution, with an average of 5 calls per hour.

Question 3a

Calculate the probability that exactly 4 calls are received in an hour and least two calls are received in an hour?

(5 marks)

Question 3b

Over a 2-hour period, what is the probability that exactly 8 calls are received?

(5 marks)

Question 3c

The hospital monitors 10 independent 1-hour periods, what is the probability that exactly 3 of these periods receive exactly 4 calls?

(5 marks)

Question 4

Let X be a continuous random variable with probability density function:

$$f(x) = \begin{cases} c(1-x^2)\,, |x| < 1 \\ 0\,, \text{otherwise}\,. \end{cases}$$

Question 4a

What is the value of c?

(5 marks)

Question 4b

Find $E(X)$ and $Var(X)$.

(10 marks)

Question 4c

Determine the cumulative distribution function F defined by

$$F(x) = P\{X \leq x\}\,,\ -\infty < x < \infty\,.$$

(5 marks)

Question 5

The time taken for a machine brand X produced by company A to complete a task follows a normal distribution with a mean of 10 minutes and a standard deviation of 4 minutes.

Question 5a

Calculate the probability that the task completion time is less than 7 minutes.

(5 marks)

Question 5b

Determine the probability that the task completion time is between 8 and 15 minutes.

(5 marks)

Question 5c

Determine the threshold time such that 30% of machines will complete the task in less than this time.

(5 marks)

----- END OF PAPER -----